# Architecture Seminar: ASL Translation

Ashley Wu, Brent Garey, Bhargav Singapuri, Paschal Corrigan

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## 1 Experiment Card

| Field | Description | Draws on |
|---|---|---|
| 1. Feature | Input: Pose keypoints, hand and face crops output by MediaPipe (We will investigate other pose estimation tools such as OpenPose as suggested in class). Output: English words representing the English translation of the signed ASL. Acts on it: The ASL signer, who may retry or eventually type messages. Any recipient of the translation who may act on the communication. | Specification Section 2 |
| 2. Task type | Classification, Sequence to Sequence Translation | Specification Section 2 |
| 3. Architectures compared | Baseline: Pose GRU Encoder, Transformer Decoder. We train the full model ourselves for comparison using a simpler encoder architecture suggested by the originator. A transformer is chosen not for popularity, but for reducing the potential for vanishing/exploding gradients during backpropagation and being able to parallelize training. Classical (Proposal): Pose GRU Encoder, RNN Decoder. Using RNNs for sequence to sequence translation is a more traditional method despite still including neural nets, we propose this over our fallback to maintain true sequence to sequence generation over classification. Classical (Fallback): Hidden Markov Model with a Sentence Bank: This is a true classical model, but would be restricting as HMMs cannot reliably generate freeform english as the classical proposal would. Deep/Foundation: Pretrained pipeline using DINOv2 for feature extraction, SHuBert for encoding, and ByT5 for encoding and decoding. This includes a fusion layer and fine tuning ByT5. Using pretrained checkpoints from models trained on large datasets greatly improves our chance at reliable translation. | Architecture Seminar Presentation |

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| 4. Split design | We will be using How2Sign's predefined train, validation, and test splits (Roughly 88:5:7) to ensure we have no leakage between subsets. How2Sign splits only include full video samples with no partials or overlaps between splits, preventing leakage. Within each split we will apply filters to impose a reduced vocabulary $\tilde{1}$000 words for one setting (Self care, fitness, food and drinks). Signers are shared between splits which is an acknowledged source of data leakage. | Specification Section 3; Datasheet. Modified to use a reduced vocabulary rather than a 10% random subset after feedback from the originator. |
| 5. Controlled conditions | The same splits, preprocessing, and metrics for every architecture. An equal tuning budget, tuned on validation data only. The test set used once. | Assignment specification, the controlled conditions example directly applys to our project. |
| 6. Metric and target | BLEURT: A metric for quantifying meaning between two sentences. Confident-Error Rate: Fraction of displayed captions (above the confidence threshold) that are incorrect. Targets: BLEURT ¿= 0.35 on test, confident-error rate ¡= 10% on the How2Sign sentences with a constrained vocabulary. | Specification Section 4; Storyboard B; Datasheet. Added confident error rate to supply a metric for the confidence signifier. |
| 7. Storyboard B slice | Test sentences with words or finger spellings that do not exist in our restricted vocabulary. | |
| 8. Calibration | Our interface shows a confidence score with each caption: the exponential of the average log-probability of its words, calibrated on the validation set. We report expected calibration error (ECE) for every architecture, computed by grouping captions into confidence bins (e.g. 0–10%, 10–20%, . . . ), taking the gap between each bin's average confidence and its fraction of correct captions, and averaging those gaps weighted by bin size. A low ECE means a caption shown at 70% confidence is correct about 70% of the time. | Architecture Seminar Presentation |
| 9. Error analysis | We will check the top 100 most frequently misclassified words in vocabulary, checked against the confidence score in Storyboard B | Storyboard B |
| 10. Decision rule | We will select the architecture with the highest BLEURT on the Storyboard B slice, among architectures that meet our targets. This incorporates our calibration into our decision rather than using the metric alone. | Architecture Seminar Presentation |
| 11. Fallback | If no architectures meet the decision rule, we will higher the confidence threshold. All other sentences will signify to the user that the sentence was not understood, showing the output with the confidence score. | Architecture Seminar |

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| 12. Technical debt | The entire pipeline relies on media pipe which is a noted limitation of our design (entanglement). Our pipeline uses frozen models which may have depended on a previous media pipe version, so we will pin versions and checkpoints (external dependencies). We are training on offline clips, and testing on a live stream, this is why we share one preprocessing module (train/test skew). We don't collect user data as recordings are very intrusive (no feedback loop). | Architecture Seminar |
| 13. Scoping conditions | As specified in the assignment specification, our scoping condidtions have now been satisfied | Assignment specification |
| 14. Labeling Plan | We will not be collecting or labeling any data ourselves. | Assignment Specification |

# 2 Comparison Table

| **Measure** | **Baseline**<br>Pose GRU encoder, Transformer decoder<br>Owner: Brent Garey | **Classical**<br>Pose GRU encoder, RNN decoder<br>Owner: Ashley Wu | **Deep or Foundation**<br>Pretrained pipeline (DINOv2, SHuBERT, fusion layer, ByT5)<br>Owners: Bhargav Singapuri, Paschal Corrigan |
|---|---|---|---|
| Overall metric<br>(BLEURT) | | | |
| Storyboard B slice<br>(BLEURT) | | | |
| Spread across seeds<br>(± std, 3 seeds) | | | |
| Calibration<br>(ECE described in Field 8) | | | |
| Training cost<br>(A100 GPU-hours) | | | |
| Inference latency<br>(p90 seconds per sentence) | | | |
| Inspectability<br>(low / medium / high) | | | |

# 3 Person-with-AI Check

To determine whether or not our product is useful we will:

1. Meet with Northeastern's local ASL community and give a walkthrough demonstration of our AI feature.
2. We will receive feedback from the ASL community on whether the product is useful and changes we can make to improve our product.